EE4-17
EE9-AO4

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

Corrected copy

MSc and EEE PART IV: MEng and ACGI

HIGH PERFORMANCE ANALOGUE ELECTRONICS

Friday, 10 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A.S. Holmes

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The Questions

1. (a) Derive an equation for the Noise Factor of a receiver, as a function of the received noise power and the equivalent input-referred noise of the system.

[4]

(b) Derive an equation for the Sensitivity of a receiver as a function of the Noise Factor, the received noise power, the Noise Figure and the Signal to Noise Ratio of the detector.

[4]

(c) Find the Noise Factor of a cascade of four circuit blocks, each with consecutive Noise Factors equal to 5, 10, 5, 10, and respective power gains equal to 1, 10, 100, 100.

[4]

(d) Assuming that there were no other constraints, how would you have chosen to cascade the blocks in (c) to optimise the noise performance? Briefly explain why.

[4]

(e) An RF signal is centred at 2.4GHz and has a bandwidth of 200MHz. Would a subsampling receiver with a sampling frequency of 250MHz be a suitable choice to receive the signal? Briefly justify your answer.

[2]

(f) A 2.4GHz transceiver operating in the receiver mode shows a large spurious signal at 4.4GHz. The intermediate frequency of the receiver is 200MHz. Which electronic block is likely to be causing the spurious signal? Justify your answer.

[2]

2. (a) Draw a CMOS differential pair transconductor with a current mirror as a load.

[2]

(b) For the circuit in (a) draw the most significant noise sources and briefly explain why those are significant.

[4]

(c) Give an expression for the equivalent output noise of the circuit in (a), as a function of the power spectral density of the noise sources you drew. Assume that transistors which play equivalent roles are matched.

[4]

(d) Draw a modified version of the circuit in (a) in which, by adding resistors, an improved linearity is achieved.

[4]

(e) Give an expression for the equivalent input noise of the circuit in (d).

[4]

(f) What is the main disadvantage of the circuit in (d) with respect to the circuit in (a)?

[2]

3. (a) Draw the schematics of CMOS transconductors with one, two and four transistors (excluding devices used for biasing).

[2]

(b) Explain in which way increasing the number of devices might improve performance.

[2]

(c) Briefly explain why matching is important in the topologies you drew in (a).

[4]

(d) Could any of the topologies you drew in (a) be used as a mixer? If so, explain how.

[4]

(e) Derive expressions for the output current and transconductance of the circuit with four transistors you drew in (a).

[4]

(f) Could any of those topologies be used as a mixer for RF frequencies around 2.4GHz, if the transistors were operating in the weak inversion operating region? Briefly justify your answer.

[4]

4. (a) A MOS transistor in a circuit is operating in a frequency band from 100kHz to 10MHz. Give an expression for the dominant sources of noise at the input.

[2]

(b) The transistor in (a) is used at the input of a transconductor differential pair, with a total Gm of 200mA/V and a 100pF capacitor at the output. Sketch the power spectral density of noise, as a function of frequency, from 10kHz to 100MHz.

[2]

(c) Would a filter built with transconductors such as the one in (b) be suitable if the input signal had a 0.3V range? Briefly explain your answer.

[4]

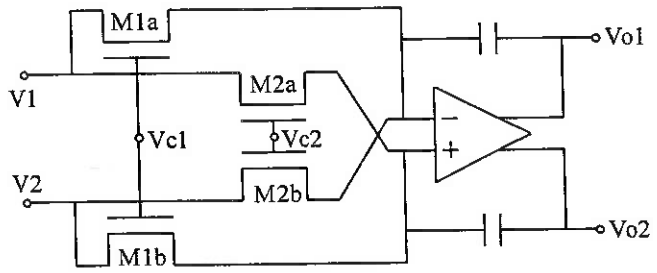
(d) How would you propose to change the blocks in the filter in (c) so that it is suitable for input swings of 0.3V.

[4]

(e) In the circuits above which intercept point should you track more carefully during the design? Briefly explain why.

[4]

(f) Explain the advantages of having a cross-coupled topology in the following circuit. What type of circuit is it?



[4]